Q23 | Correct option: A

Source: Hureau et al., Journal of Physiology, On the role of skeletal muscle acidosis and inorganic phosphates — PDF page 23 of 30; 'KEY POINTS', final bullet and preceding measurements bullet

KEY POINTS.

- We investigated the relationship between intramuscular metabolites and neuromuscular function in humans performing 2 maximal, intermittent, knee-extension trials interspersed with 5min of rest.
- Concomitant measurements of intramuscular hydrogen (H^+) and inorganic phosphate (Pi) concentrations, and quadriceps twitch-force (Q_{tw}) and voluntary activation (VA), were made throughout each trial using ^{31}P-MRS and electrical femoral nerve stimulations.
- While [Pi] fully recovered prior to the onset of the second trial, [H^+] did not.
- Q_{tw} was strongly related to both [H^+] and [Pi] across both trials. However, the relationship between Q_{tw} and [H^+] shifted leftward from the first to the second trial, while the relationship between Q_{tw} and [Pi] remained unaltered. VA was related to [H^+], but not [Pi], across both trials.
- These *in-vivo* findings support the hypotheses of intramuscular Pi as a primary cause of peripheral fatigue, and muscle acidosis, likely acting on group III/IV muscle afferents, as a contributor to central fatigue.

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Biographies

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